विशाल जाधव सरांचे

VJTech Academy

Inspiring Your Success...

Program: Unit - I Basic Syntactical Constructs in Java

//Write a Java program to print "Welcome" message on output screen.

```
class Welcome
{
	public static void main(String args[])
	{
		System.out.println("Welcome to VJTech Academy")
	}
}
```

//Write a Java program to perform addition between two numbers.

```
class Addition
{
	public static void main(String args[])
	{
		int a,b,c;
		a=100;
		b=50;
		c=a+b;
		System.out.println("Addition="+c);
	}
}
```

//Command Line Arguments

```
class CommandLineArgs
{
	public static void main(String args[])
	{
		int a,b,c;
		a=Integer.parseInt(args[0]);
		b=Integer.parseInt(args[1]);
		c=a+b;
		System.out.println("Addition="+c);
	}
}
```

//Scanner class - Used to take input from user

```
import java.util.*;
class ScannerDemo
{
	public static void main(String args[])
	{
		int a,b,c;
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter 1st Number:");
		a=sc.nextInt();
		System.out.println("Enter 2nd Number:");
		b=sc.nextInt();
		c=a+b;
		System.out.println("Addition="+c);
	}
}
```

//Math Class

```
class MathDemo
{
	public static void main(String args[])
	{
		System.out.println("Minimum Value="+Math.min(10,20));
		System.out.println("Maximum Value="+Math.max(10,20));
		System.out.println("Square Root of 9="+Math.sqrt(9));
		System.out.println("pow(2,3)="+Math.pow(2,3));
		System.out.println("Exponential of 709.78222656 is
"+Math.exp(709.78222656));
		System.out.println("round(260.57)="+Math.round(260.57));
		System.out.println("Absolute Value = "+Math.abs(-5944));
	}
}
```

OUTPUT
========
Minimum Value=10
Maximum Value=20
Square Root of 9=3.0
pow(2,3)=8.0
Exponential of 709.78222656 is 1.7968190692375724E308
round(260.57)=261
Absolute Value = 5944

//final keyword (Constant Declaration)

```
import java.util.*;
class FinalKeywordDemo
{
	public static void main(String args[])
	{
		Scanner sc=new Scanner(System.in);
		float radius,area;
		final float PI=3.14f;
		System.out.println("Enter radius of circle:");
		radius=sc.nextFloat();
		area=(PI*radius*radius);
		System.out.println("Area of circle="+area);
	}
}
OUTPUT
========
Enter radius of circle:
2.5
Area of circle=19.625
```

//Implicit Type casting

```
class ImplicitTypeCastingDemo
{
        public static void main(String args[])
        {
            int a=70;
            float b;
            b=a;
            System.out.println("Value of int variable a="+a);
            System.out.println("Value of float variable b="+b);
        }
}
```

//Explicit Type casting

```
class ExplicitTypeCastingDemo
{
        public static void main(String args[])
        {
            int a=70;
            float b;
            b=(float)a;
            System.out.println("Value of int variable a="+a);
            System.out.println("Value of float variable b="+b);
        }
}
```

//if statement

```
import java.util.*;
class IfStatement
{
	public static void main(String args[])
	{
		Scanner sc=new Scanner(System.in);
		int a,b;
		System.out.println("Enter two numbers:");
		a=sc.nextInt();
		b=sc.nextInt();
		if(a==b)
		{
			System.out.println("Both numbers are equal");
		}

	}
}
```

//if else statement

```
import java.util.*;
class IfElseStatement
{
	public static void main(String args[])
	{
		Scanner sc=new Scanner(System.in);
		int a,b;
		System.out.println("Enter two numbers:");
		a=sc.nextInt();
		b=sc.nextInt();
		if(a==b)
		{
			System.out.println("Both numbers are equal");
		}
		else
		{
			System.out.println("Both numbers are not equal");

		}

	}
}
```

//if else statement

```
import java.util.*;
class CheckEvenOdd
{
	public static void main(String args[])
	{
		Scanner sc=new Scanner(System.in);
		int no;
		System.out.println("Enter any number:");
		no=sc.nextInt();
		if(no%2==0)
		{
			System.out.println("Even Number");
		}
		else
		{
			System.out.println("Odd Number");
		}

	}
}
```

//instanceof operator

```
import java.util.*;
class InstanceOfOpDemo
{
	public static void main(String args[])
	{
		Scanner sc=new Scanner(System.in);
		if(sc instanceof Scanner)
		{
			System.out.println("sc is an object of Scanner class");
		}
	}
}
```

//Nested if else statement

```
import java.util.*;
class NestedIfElse
{
	public static void main(String args[])
	{
		Scanner sc=new Scanner(System.in);
		int no;
		System.out.println("Enter any number:");
		no=sc.nextInt();
		if(no!=0)
		{
			if(no>0)
			{
				System.out.println("Number is positive");
			}
			else
			{
				System.out.println("Number is negative");
			}
		}
		else
		{
			System.out.println("Zero is neigher positive nor negative");
		}
	}
}
```

//else if ladder statement

```
import java.util.*;
class ElseIfLadder
{
	public static void main(String args[])
	{
		Scanner sc=new Scanner(System.in);
		int marks;
		System.out.println("Enter your marks:");
		marks=sc.nextInt();
		if(marks>=75)
		{
			System.out.println("You got distinction");
		}
		else if(marks>=60)
		{
			System.out.println("You got first class");
		}
		else if(marks>=40)
		{
			System.out.println("You are pass only");
		}
		else
		{
			System.out.println("You are fail");
		}

	}
}
```

//switch case statement

```
import java.util.*;
class SwitchCaseStmt
{
	public static void main(String args[])
	{
		Scanner sc=new Scanner(System.in);
		int no;
		System.out.println("Enter any number:");
		no=sc.nextInt();
		switch(no)
		{
			case 1: System.out.println("ONE");
				break;
			case 2: System.out.println("TWO");
				break;
			case 3: System.out.println("THREE");
				break;
			case 4: System.out.println("FOUR");
				break;
			default:System.out.println("Enter no between 1 to 4");
				break;
		}
	}
}
```

//for loop

```
class forloopDemo
{
	public static void main(String args[])
	{
		int i;
		for(i=1;i<=5;i++)
		{
			System.out.println("VJTech Academy");
		}
	}
}
```

//while loop

```
class whileloopDemo
{
	public static void main(String args[])
	{
		int i;
		i=1;
		while(i<=5)
		{
			System.out.println("VJTech Academy");
			i++;
		}
	}
}
```

//do-while loop

```
class DowhileloopDemo
{
	public static void main(String args[])
	{
		int i;
		i=1;
		do
		{
			System.out.println("VJTech Academy");
			i++;
		}while(i<=5);
	}
}
```

//for each loop

```
class ForEachLoop
{
	public static void main(String args[])
	{
		int num[]={10,20,30,40,50};
		for(int x:num)
		{
			System.out.println("Value of x="+x);
		}
	}
}
OUTPUT
======
Value of x=10
Value of x=20
Value of x=30
Value of x=40
Value of x=50
```

//WAP to perform addition between two numbers using class and object

```
class Addition
{
	int a,b,c;
	void getdata()
	{
		a=100;
		b=200;
	}
	void display()
	{
		c=a+b;
		System.out.println("Addition="+c);
	}
	public static void main(String args[])
	{
		Addition a1=new Addition();
		a1.getdata();
		a1.display();
	}
}
```

WAP to create class Student having data members rollno,name and marks.Accept and display data for one student.

```
import java.util.*;
class Student
{
	int rollno;
	String name;
	float marks;
	void get_stud_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Roll No:");
		rollno=sc.nextInt();
		System.out.println("Enter Name:");
		name=sc.next();
		System.out.println("Enter Marks:");
		marks=sc.nextFloat();
	}
	void disp_stud_info()
	{
		System.out.println("******Student Info*******");
		System.out.println("Roll No:"+rollno);
		System.out.println("Name:"+name);
		System.out.println("Marks:"+marks);
	}
	public static void main(String args[])
	{
		Student s1=new Student();
		s1.get_stud_info();
		s1.disp_stud_info();
	}
}
/*
```

```
OUTPUT
======
Enter Roll No:
1010
Enter Name:
James
Enter Marks:
98.90
******Student Info*******
Roll No:1010
Name:James
Marks:98.9
*/
```

WAP to create class Book having data members bookid,book name,Author Name and price.Accept and display data for two books.

```
import java.util.*;
class Book
{
	int bookid;
	String book_name,author_name;
	float price;
	void get_book_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Book ID:");
		bookid=sc.nextInt();
		System.out.println("Enter Book Name:");
		book_name=sc.next();
		System.out.println("Enter Author Name:");
		author_name=sc.next();
		System.out.println("Enter Book Price:");
		price=sc.nextFloat();
	}
	void disp_book_info()
	{
		System.out.println("******Book Info*******");
		System.out.println("Book ID:"+bookid);
		System.out.println("Book Name:"+book_name);
		System.out.println("Author Name:"+author_name);
		System.out.println("Book Price:"+price);
	}
	public static void main(String args[])
	{
		Book b1=new Book();
		Book b2=new Book();
		b1.get_book_info();
		b2.get_book_info();
		b1.disp_book_info();
```

```
                b2.disp_book_info();
        }
}
OUTPUT
======
Enter Book ID:
1111
Enter Book Name:
Java
Enter Author Name:
James
Enter Book Price:
780
Enter Book ID:
2222
Enter Book Name:
Python
Enter Author Name:
Brenden
Enter Book Price:
950
******Book Info*******
Book ID:1111
Book Name:Java
Author Name:James
Book Price:780.0
******Book Info*******
Book ID:2222
Book Name:Python
Author Name:Brenden
Book Price:950.0
```

Constructor Overloading

```
class Item
{
        int x;
        Item()//default constructor
        {
                x=100;
        }
        Item(int p)   //Parameterized constructor
        {
                x=p;
        }
        Item(Item m)        //Copy constructor
        {
                x=m.x;
        }
        void display()
        {
                System.out.println("Value of X : "+x);
        }
        public static void main(String args[])
        {
                 Item i1=new Item();
                 Item i2=new Item(i1);
                 Item i3=new Item(570);
                 i1.display();
                 i2.display();
                 i3.display();
        }
}
```

One Dimensional Array

```
class OneDArray
{
	public static void main(String args[])
	{
		int a[]=new int[5];
		a[0]=10;
		a[1]=20;
		a[2]=30;
		a[3]=40;
		a[4]=50;
		for(int i=0;i<a.length;i++)
		{
			System.out.println("Element present at index "+i+" is "+a[i]);
		}
	}
}
Element present at index 0 is 10
Element present at index 1 is 20
Element present at index 2 is 30
Element present at index 3 is 40
Element present at index 4 is 50
```

One Dimensional Array

```
import java.util.*;
class OneDArray1
{
	public static void main(String args[])
	{
		int a[]=new int[5];
		int i;
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Five Array Elements:");
		for(i=0;i<5;i++)
		{
			a[i]=sc.nextInt();
		}
		System.out.println("Array Elements are:");
		for(i=0;i<5;i++)
		{
			System.out.print(a[i]+" ");
		}
	}
}
Enter Five Array Elements:
100
200
300
400
500
Array Elements are:
100 200 300 400 500
```

One Dimensional Array

```
import java.util.*;
class OneDArray2
{
	public static void main(String args[])
	{
		int a[]=new int[5];
		int i;
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Five Array Elements:");
		for(i=0;i<5;i++)
		{
			a[i]=sc.nextInt();
		}
		System.out.println("Array Elements in reverse order:");
		for(i=4;i>=0;i--)
		{
			System.out.print(a[i]+" ");
		}
	}
}
Enter Five Array Elements:
100
200
300
400
500
Array Elements in reverse order:
500 400 300 200 100
```

Two Dimensional Array

```
import java.util.*;
class TwoDArray
{
                public static void main(String args[])
                {
                    int a[][]=new int[3][3];
                    int i,j;
                    Scanner sc=new Scanner(System.in);
                    System.out.println("Enter 3*3 Array Elements:");
                    for(i=0;i<3;i++)
                    {
                        for(j=0;j<3;j++)
                        {
                            a[i][j]=sc.nextInt();
                        }
                    }
                    System.out.println("Array Elements are:");
                    for(i=0;i<3;i++)
                    {
                        for(j=0;j<3;j++)
                        {
                            System.out.print(a[i][j]+" ");
                        }
                        System.out.println();
                    }
                }
}
/*
Enter 3*3 Array Elements:
10
20
30
40
50
```

```
60
70
80
90
Array Elements are:
10 20 30
40 50 60
70 80 90
*/
```

Write a program to create vector with six elements (10,30,60,70,80,100). Removes element 3rd and 4th position. Insert new element at 3rd position. Display the original and current size of vector

```
import java.util.*;
class VectorDemo1
{
	public static void main(String args[])
	{
		Vector v1=new Vector();
		v1.addElement(new Integer(10));
		v1.addElement(new Integer(30));
		v1.addElement(new Integer(60));
		v1.addElement(new Integer(70));
		v1.addElement(new Integer(80));
		v1.addElement(new Integer(100));
		System.out.println("Initial Vector Size = "+v1.size());
		v1.removeElementAt(3);
		v1.removeElementAt(4);
		v1.insertElementAt(new Integer(150),3);
		System.out.println("Final Vector Size = "+v1.size());
	}
}
```

Vector Class

```
import java.util.*;
class VectorDemo2
{
	public static void main(String args[])
	{
		Vector v1=new Vector();
		for(int i=0;i<args.length;i++)
		{
			v1.addElement(args[i]);
		}
		System.out.println("Vector of Size = "+v1.size());
		System.out.println("Your Vector Elements:");
		for(int i=0;i<v1.size();i++)
		{
			System.out.println(v1.elementAt(i));
		}
	}
}
```

Wrapper Class

```
class WrapperClassDemo
{
	public static void main(String args[])
	{
		int i=100;
		Integer ii=new Integer(i);
		System.out.println("Primitive Integer Value = "+i);
		System.out.println("Object Integer Value = "+ii);
	}
}
/*
Primitive Integer Value = 100
Object Integer Value = 100
*/
```

//String class

```
class StringDemo
{
    public static void main(String args[])
    {
       String str=new String("VJTech");
       System.out.println("Value of str ="+str);
       System.out.println("Length of str ="+str.length());
       System.out.println("To Lower Case ="+str.toLowerCase());
       System.out.println("To Upper Case ="+str.toUpperCase());
       System.out.println("Character present at 2 index ="+str.charAt(2));
       System.out.println("Concatenation="+str.concat("Academy"));
       System.out.println("Index of e character ="+str.indexOf('e'));
       System.out.println("Equals method ="+str.equals("VJTech"));
       System.out.println("CompareTo method ="+str.compareTo("VJTech"));
    }
}
```

OUTPUT:

```
=======
Value of str =VJTech
Length of str =6
To Lower Case =vjtech
To Upper Case =VJTECH
Character present at 2 index =T
Concatenation=VJTechAcademy
Index of e character =3
Equals method =true
CompareTo method =0
```